Infinite NPA depth at minimal semidefinite complexity

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The quantum region of the CHSH value and the sum of two marginal averages requires infinite Navascués–Pironio–Acín depth, yet admits an exact finite semidefinite lift. We construct this lift by showing that every supporting bound is attained with equal measurement overlaps: one angle determines the entire boundary. Finite NPA certificates must remain valid for two independently varying angles, where the obstruction to finite depth survives. The same region has the smallest semidefinite extension degree possible for a nonpolyhedral set, separating infinite hierarchy depth from minimal semidefinite complexity.

Alice and Bob each choose between two measurements with outcomes ±1. Write $a_x = \langle A_x\rangle$, $b_y = \langle B_y\rangle$ and $E_{xy} = \langle A_xB_y\rangle$, and retain

$$M = a_0 + b_0, \qquad S = E_{00} + E_{01} + E_{10} - E_{11}. \quad (1)$$

S is the CHSH value [1]. Let $\mathcal{W}$ be their full quantum region. The two marginal averages may differ; only their sum is retained. Mixing strategies makes $\mathcal{W}$ convex, and the two-qubit reduction below makes it compact.

The preceding Letter [2] proved that every finite NPA level [3, 4] overestimates some supporting lines $S + \alpha M$ near $\alpha = 1$. Here we construct an exact finite lift of this same region. Its semidefinite extension degree—the smallest possible largest PSD block order in a finite lift—is two:

$$\mathcal{W} \subsetneq \mathcal{W}_L \quad \text{for every finite } L, \qquad \mathrm{sxdeg}(\mathcal{W}) = 2. \quad (2)$$

The explicit construction below uses blocks of orders five and four. The optimal order two follows from the geometry of the region.

Equal overlaps determine every supporting line.— To find $\mathcal{W}$, fix a candidate inequality $hS + \alpha M \le \lambda$. For given measurements its validity in every state is equivalent to

$$\lambda I - H \succeq 0, \qquad H = h\,\mathrm{CHSH} + \alpha(A_0 + B_0).$$

After projective dilation, Jordan's lemma decomposes each pair of binary observables into blocks of dimension at most two. Choosing local axes within these blocks gives

$$\begin{aligned} A_0 = B_0 = Z, \\ A_1 = cZ + \sqrt{1-c^2}X, \quad B_1 = dZ + \sqrt{1-d^2}X. \end{aligned} \quad (3)$$

The cosines $c, d \in [-1, 1]$ are the local measurement overlaps. Deterministic strategies attain $\ell = 2|h| + 2|\alpha|$, so a valid bound must satisfy $\lambda \ge \ell$.

Above this bound, the determinant decides positivity of the slack. To see this, put $\Omega = 1 + c + d - cd \in [-2, 2]$ and expand $\det(rI + \lambda I - H) = r^4 + e_1r^3 + e_2r^2 + e_3r + D$:

$$\begin{aligned} e_1 = 4\lambda, \qquad e_2 = 6\lambda^2 - 4\alpha^2 - 8h^2, \\ e_3 = 4\lambda^3 - 8\lambda\alpha^2 - 16\lambda h^2 - 8h\alpha^2\Omega. \end{aligned} \quad (4)$$

These three coefficients are nonnegative. For e_3, replacing its last term by $-16|h|\alpha^2$ gives an increasing cubic on $\lambda \ge \ell$, whose value at ℓ is $16|\alpha|(|\alpha| + 2|h|)^2$. If $D \ge 0$ too, the polynomial is positive for every $r > 0$; a negative eigenvalue of the slack would give such a root.

We can therefore compare measurement choices through a scalar. Set $x = (1 - c)/2$, $y = (1 - d)/2$, and write $D(x, y) = \det(\lambda I - H)$. Direct expansion gives

$$D(x, y) = F(xy) + 64h^2(4h^2 - \alpha^2)xy(x + y), \quad (5)$$

where F depends only on the product. For $|\alpha| \le |h|$, the coefficient of the sum is nonnegative. At fixed product, $x + y \ge 2\sqrt{xy}$, so

$$D(x, y) \ge D(\sqrt{xy}, \sqrt{xy}). \quad (6)$$

The equal-overlap point remains in $[0, 1]^2$. Positivity there implies positivity at the original pair. Thus equal overlaps are the hardest measurements on which to satisfy the supporting bound.

For $|\alpha| \ge |h|$, the support is simply ℓ. The endpoint tilted-CHSH inequality follows from the identity

$$\begin{aligned} 4I - (\mathrm{CHSH} + 2A_0) = (I - C - A_1D_B)^2 \\ + \tfrac{1}{4}\sum_{y=0}^{1}[(I - A_0)(I + B_y)]^2, \end{aligned} \quad (7)$$

where $C = (B_0 + B_1)/2$ and $D_B = (B_0 - B_1)/2$. Party exchange and output flips give either sign of CHSH and either sign of the marginal. Average the Alice and Bob bounds, multiply by $|h|$, and add the remaining marginal slack of weight $|\alpha| - |h|$. This proves $H \le \ell I$, attained with equal-overlap deterministic measurements. Equal overlaps therefore suffice for every support direction.

One angle gives one finite lift.— At $c = d$, exchange symmetry separates the singlet, whose eigenvalue is $-2hc$. On the other three Bell states, set $c = (1 - t^2)/(1 + t^2)$ and multiply the slack by $(1 + t^2)^2$. This removes the square root and gives

$$K(t) = \begin{pmatrix} \lambda p_0 - 2hp_1 & -2\alpha p_0 & 0 \\ -2\alpha p_0 & \lambda p_0 - 2hp_2 & -8ht^3 \\ 0 & -8ht^3 & \lambda p_0 + 2hp_2 \end{pmatrix}, \quad (8)$$

with $p_0 = (1+t^2)^2$, $p_1 = 1-t^4$ and $p_2 = 1+4t^2-t^4$. At $t=0$ and in the leading coefficient, positivity forces $\lambda \geq 2h + 2|\alpha|$ and $\lambda \geq -2h + 2|\alpha|$. These imply $\lambda \geq \ell$ and control the singlet. Continuity includes $c = -1$, while $K(-t) = RK(t)R$, $R = \mathrm{diag}(1,1,-1)$. Consequently, for all real support coefficients,

$$hS + \alpha M \leq \lambda \text{ on } \mathcal{W} \quad \Longleftrightarrow \quad K(t) \succeq 0 \quad \forall t \in \mathbb{R}. \tag{9}$$

The infinitely many angle tests now have a finite certificate. A PSD real univariate matrix polynomial is a polynomial matrix sum of squares, including when singular [6]. Quartic diagonal entries force every factor to have degree at most two, since leading real squares cannot cancel. Thus the same finite feature list works for every supporting line. For $\xi \in \mathbb{R}^3$, parity separates it into

$$Z_+ = (\xi_1, \xi_2, t\xi_3, t^2\xi_1, t^2\xi_2)^T,$$
$$Z_- = (\xi_3, t\xi_1, t\xi_2, t^2\xi_3)^T.$$

Equation (9) is therefore equivalent to the polynomial identity

$$\xi^T K(t) \xi = Z_+^T G Z_+ + Z_-^T J Z_-, \qquad G, J \succeq 0. \tag{10}$$

Matching the sixteen independent coefficients defines a linear map $\mathcal{A}(G,J) = Bq$, where $q = (\lambda, \alpha, h)$. Because K is affine in all three entries of q, this describes the entire cone of valid inequalities in one SDP.

To recover the region, take the adjoint maps

$$(U(z), V(z)) = \mathcal{A}^* z, \qquad B^T z = (\ell(z), -m(z), -s(z)).$$

Here $z \in \mathbb{R}^{16}$, and U, V have orders five and four. A feasible z with $\ell(z) = 1$ satisfies

$$\lambda - \alpha m(z) - h s(z) = \mathrm{tr}(UG) + \mathrm{tr}(VJ) \geq 0 \tag{11}$$

for every valid supporting bound. Thus $(m(z), s(z)) \in \mathcal{W}$.

For the converse, the positive definite pair

$$G_0 = \mathrm{diag}(1,1,2,1,1), \qquad J_0 = \mathrm{diag}(1,2,2,1)$$

represents $q = (1,0,0)$ and supplies strict feasibility. Moreover, $\ell(z) = \mathrm{tr}(U(z)G_0) + \mathrm{tr}(V(z)J_0)$. A bounded normalization therefore bounds both PSD matrices. Every coordinate of z occurs in their entries, so the feasible auxiliaries are bounded too. The dual image is closed, and semidefinite duality gives the reverse inclusion:

$$\mathcal{W} = \{(m(z), s(z)) : \ell(z) = 1,\ U(z), V(z) \succeq 0\}. \tag{12}$$

This is the exact finite lift, with every boundary point attained. The coefficient equations and rational matrices are listed in the complete proofs [5].

Infinite depth, minimal block order.— The equal-overlap reduction concerns the support values. Finite NPA certificates must hold as observable-word identities for independent measurement angles. Near $\alpha = 1$, our earlier proof uses this two-variable freedom to produce a shifted Motzkin polynomial that cannot be a sum of squares [2].

For $f(\alpha) = \max_{\mathcal{W}}(S + \alpha M)$, the quantum boundary satisfies [2, 7]

$$f(1-T) = 4 - 2T + \tfrac{4}{3}T^3 + O(T^4), \qquad T \downarrow 0.$$

A polygon would have affine support sufficiently near this direction, so $\mathcal{W}$ is nonpolyhedral. Its finite lift makes it semialgebraic. Scheiderer's theorem gives extension degree at most two for every closed convex semialgebraic set in the plane [8]; order-one lifts are polyhedral. Hence $\mathrm{sxdeg}(\mathcal{W}) = 2$.

The construction also gives thirty-two exact projections. For each CHSH sign pattern χ with $\prod_{ij} \chi_{ij} = -1$, and each selected input pair x, y, physical relabellings identify

$$\left(a_x + \chi_{xy} b_y,\ \sum_{ij} \chi_{ij} E_{ij} \right)$$

with the same region $\mathcal{W}$. Adding these lifts to any NPA level retains every quantum correlation and makes all supporting bounds on these planes exact, including the family responsible for finite-level nontermination.

[1] J. F. Clauser, M. A. Horne, A. Shimony, and R. A. Holt, Proposed experiment to test local hidden-variable theories, Phys. Rev. Lett. **23**, 880 (1969).

[2] A. Chaturvedi, *No Finite NPA Level Characterizes the Complete Quantum Set in the Simplest Bell Scenario*, manuscript (2026).

[3] M. Navascués, S. Pironio, and A. Acín, Bounding the set of quantum correlations, Phys. Rev. Lett. **98**, 010401 (2007).

[4] M. Navascués, S. Pironio, and A. Acín, A convergent hierarchy of semidefinite programs characterizing the set of quantum correlations, New J. Phys. **10**, 073013 (2008).

[5] See the accompanying Supplemental Material for the complete finite-lift proof, explicit membership matrices, all signed asymmetric support branches, physical relabellings and the full earlier NPA proof.

[6] C. Hanselka and R. Sinn, Positive semidefinite univariate matrix polynomials, Math. Z. **292**, 83–101 (2019), Corollary 2.8.

[7] N. Gigena, E. Panwar, G. Scala, M. Araújo, M. Farkas, and A. Chaturvedi, Self-testing tilted strategies for maximal loophole-free nonlocality, npj Quantum Inf. **11**, 82 (2025).

[8] C. Scheiderer, Second-order cone representation for convex sets in the plane, SIAM J. Appl. Algebra Geom. **5**, 114–139 (2021), Theorem 0.1.